Structural Inevitability of the Einstein Constraints

Sina Montazeri
University of North Texas
sinamontazeri@my.unt.edu

Abstract

The Einstein constraints are structurally inevitable. On a smooth manifold, tangent spaces globalize to bundles, bundles acquire connections, and connections carry curvature. When the geometry admits a tangent-normal splitting, that curvature decomposes into Gauss, mixed, and Ricci channels, and the mixed channel records the failure of the pointwise splitting to remain globally aligned.

Fixing the intrinsic metric and Levi-Civita connection isolates the residual split orthogonal compatibility geometry carried by the normal connection and second fundamental form. This frozen-tangential geometry is generically mixed and non-elliptic.

In codimension one, the Ricci channel disappears, and the mixed/Gauss remnant exhausts the natural linear metric vector-scalar content, with no cross-channel output. Consequently, the codimension-one compatibility geometry is closed and determines a unique admissibility structure, which after unit-normal identification is exactly the fixed-metric vacuum Einstein constraint structure [7, 8, 9].

I Introduction

In geometric settings, structural elements and their couplings limit what further structure is admissible, sometimes to the point of uniqueness. A combination of structural constraints and their propagation makes the Einstein constraints uniquely admissible. To study this, one must adopt the point of view of a structure determining its own equilibrium. What follows lays out the necessary foundations of structural admissibility as a point of view.

Differentiability, through the simple fact that the notion of change exists, takes causal priority in this view. This principle makes a smooth manifold the first setting in which geometric change can exist. It is also the geometric setting in which that notion can be modeled precisely.

From that local structure one obtains tangent spaces pointwise, and from their global assembly one obtains bundles. At this stage, the geometric vehicle is only fiberwise, since linear data exists at each point but there is not yet a way to compare those data across the manifold.

A second fact, that the notion of variation exists, makes a connection the meaningful carrier of relation between fibers. A connection is the minimal device that allows differentiation on a bundle, and articulates its change. Curvature is therefore already present in principle, because parallel transport around a closed loop need not return a vector unchanged. Curvature is the intrinsic record of that failure.

Applying the same logic to subspaces inside each fiber, a connection preserves them, couples them, or mixes them under transport. Therefore from the notions of change and variation, decomposition becomes geometrically meaningful. A splitting is no longer just linear algebra at a point but becomes a global structural feature of the bundle and its connection. The connection can then be read relative to the distinguished subspaces that the splitting has introduced. It is read in terms of

how it preserves, couples, or mixes those subspaces, and thereby acquires a block structure adapted to the splitting.

Curvature, still carried by the connection, can be read in block form as well. It therefore decomposes into three channels according to the splitting. One channel remains within one side of the decomposition, one couples the two sides, and one remains within the other side. One of these subspaces carries directions intrinsic to the structure, while the other carries directions normal to it. This splitting is the tangent-normal decomposition of the bundle

$$E = TM \oplus N.$$

In the tangent-normal decomposition, these are the Gauss, Codazzi, and Ricci channels. These channels are the inevitable faces of one curvature once that curvature is viewed through a split geometric structure.

In other words, the mixed block records exactly the misalignment between the pointwise splitting and the global connection during transport across the manifold. It is the generic sign that a pointwise distinction has been globalized and must now be compatible with transport, differentiation, and curvature. The mixed channel is therefore where the residual compatibility geometry becomes most visible.

To study that mixed channel, the intrinsic metric h and its Levi-Civita connection ∇^h may be fixed, leaving the orthogonal compatibility geometry carried by the normal connection and the mixed data. The resulting assembled connection still carries curvature, still decomposes into channels, and still contains a mixed component. In the frozen geometry, that mixed component is no longer one term among many in the full classical system. It is where the residual geometry remains open.

In codimension one, since $\mathfrak{so}(1) = 0$, the normal-curvature channel vanishes, so only the mixed and Gauss channels remain. The resulting rank-one remnant therefore has only these two surviving sources of natural output.

Invariant theory then determines what outputs are possible. The mixed channel admits a unique natural linear metric vector contraction, the Gauss channel admits a unique scalar contraction, and there are no cross-channel outputs. Thus the codimension-one remnant is closed in the natural linear metric vector-scalar category.

These two contractions are exactly the fixed-metric vacuum Einstein constraints. The mixed contraction is

$$\nabla^j K_{ij} - \nabla_i \operatorname{tr}_h K,$$

and the scalar Gauss contraction is

$$\operatorname{scal}_h - |K|_h^2 + (\operatorname{tr}_h K)^2.$$

Therefore, the structure arising from the notions of change and variation determines its own vector-scalar constraint structure. The codimension-one admissibility structure is rigid, and after unit-normal identification its zero locus is exactly the fixed-metric vacuum Einstein constraint locus.[1]

A Main Results

The main results are as follows.

[1] The assembled connection viewpoint is classical; cf. [3, Vol. II, Ch. VII] for the Gauss–Codazzi–Ricci framework in connection form. Formal integrability theory is treated in [2].

1. The canonical connection $\nabla^{\oplus}$ on E has curvature decomposition

$$\mathcal{F}^{\oplus} = \begin{pmatrix} \mathcal{G} & -\mathcal{C}^* \\ \mathcal{C} & \mathcal{S} \end{pmatrix},$$

   and vanishing of these channels is equivalent to the Gauss–Codazzi–Ricci system (Theorems II.2 and II.4).

2. The admissible connection space $\mathcal{A}_{\oplus}$ carries a gauge-equivariant flatness operator Φ, and at flat compatible backgrounds with $K_0 = 0$ the reduced principal symbol in the mixed sector fails injectivity at every nonzero covector; no admissible internal gauge fixing restores ellipticity (Proposition III.3, Theorems IV.5 and IV.7).

3. On frozen flat domains, the mixed kernel is generated by Hessians, and the reduced formal tangent of the flat locus contains an infinite-dimensional mixed-sector subspace (Theorems V.1 and V.3).

4. In codimension one, rank one removes the Ricci channel, the surviving mixed/Gauss pair defines an intrinsic constraint map, and that map is exactly the fixed-metric vacuum Einstein constraint map after unit-normal identification (Theorems VI.1, VI.3, and VI.5).

5. In codimension one, classification and exclusion close the residual vector-scalar structure, determine the unique admissibility structure, and force the corresponding zero locus to be rigid (Theorems VI.10 and VI.11, Corollary VI.12).

II Orthogonal Geometry

Let M be a smooth manifold with fixed intrinsic metric h and Levi-Civita connection ∇^h, and let N be a metric normal bundle. The residual compatibility geometry is then carried by the split orthogonal bundle

$$E = TM \oplus N,$$

together with the orthogonal connections assembled from the fixed intrinsic block and the variable normal and mixed data. The curvature of this structure is therefore forced to decompose into Gauss, Codazzi, and Ricci channels.
To write this geometry explicitly, equip E with the product metric $g_E := h \oplus g_N$, and write

$$\nabla^{\text{prod}} := \nabla^h \oplus \nabla^{\perp}.$$

Curvature is taken with the convention

$$R(X,Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z.$$

The induced fiberwise pairing on $\mathfrak{so}(E)$ is denoted by $\langle \cdot, \cdot \rangle_{\mathfrak{so}(E)}$.

Definition II.1 (GCR data)**.** GCR data *consist of a triple* $(h, K, \nabla^{\perp})$ *where* h *is a Riemannian metric on* M*,* $K \in \Gamma(\mathrm{Sym}^2 T^*M \otimes N)$*, and* $\nabla^{\perp}$ *is a metric connection on* N*, satisfying*

$$D^{\perp}_X K(Y,Z) = \big(D^T_X K\big)(Y,Z) + R^{\perp}(X,Z)Y \qquad \forall\, X,Y,Z \in \Gamma(TM). \tag{II.1}$$

and

$$D^{\mathrm{Hom}} K = 0. \tag{II.2}$$

where $R^{\perp}$ *is the curvature of* $\nabla^{\perp}$ *and* $D^{\mathrm{Hom}}_X K := D^T_X\big(K(\cdot)\big) \in \Omega^2(M, \mathrm{Hom}(TM,N))$*.*

The split connection geometry is expressed through the covariant operators induced by ∇^h and $\nabla^\perp$, the mixed tensor K and its adjoint, and the resulting orthogonal perturbation of the product connection.
The induced covariant operators are

$$D^T : \Omega^*(M, T^*M \otimes N) \to \Omega^{*+1}(M, T^*M \otimes N), \qquad D^\perp : \Omega^*(M, N) \to \Omega^{*+1}(M, N).$$

The mixed tensor K defines bundle maps

$$K_X := K(X, \cdot) \in \Gamma(\mathrm{Hom}(TM, N)),$$

and its adjoint $K_X^* \nu \in \Gamma(TM)$ is characterized by

$$h(K_X^* \nu, Y) = \langle \nu, K_X Y \rangle_N \qquad \forall\, Y \in TM,\ \nu \in N.$$

The mixed covariant derivative is then

$$D^{\mathrm{Hom}} : \Omega^1(M, \mathrm{Hom}(TM, N)) \to \Omega^2(M, \mathrm{Hom}(TM, N)), \qquad D_X^{\mathrm{Hom}} K := D_X^T(K(\cdot)).$$

With these pieces fixed, the perturbation from the product connection is

$$\Psi_X = \begin{pmatrix} 0 & -K_X^* \\ K_X & 0 \end{pmatrix} \in \mathfrak{so}(E)_x, \qquad X \in T_x M.$$

For bundle-valued forms, $\wedge$ denotes the graded product induced by composition in $\mathrm{End}(E)$. The corresponding quadratic block identities are

$$(\Psi \wedge \Psi)_{TT} = -(K^* \wedge K), \qquad (\Psi \wedge \Psi)_{TN} = 0, \qquad (\Psi \wedge \Psi)_{NN} = -(K \wedge K^*).$$

A Canonical Assembly

Given GCR data $(h, K, \nabla^\perp)$ in the sense of Definition II.1, the split orthogonal geometry is assembled into a single metric connection

$$\nabla^\oplus = \nabla^h \oplus \nabla^\perp + \Psi(K).$$

Theorem II.2 (Canonical Assembly)**.** *Given GCR data $(h, K, \nabla^\perp)$, there exists a unique metric connection $\nabla^\oplus$ on $E = TM \oplus N$ whose tangential block is ∇^h, whose normal block is $\nabla^\perp$, and whose mixed block is K_X. Equivalently, the opposite mixed block is $-K_X^*$.*

Proof. Set

$$\nabla_X^\oplus (Y \oplus \nu) := \big(\nabla_X^h Y - K_X^* \nu\big) \oplus \big(\nabla_X^\perp \nu + K_X Y\big).$$

Equivalently, $\nabla^\oplus = \nabla^{\mathrm{prod}} + \Psi$. Since ∇^{prod} is metric and $\Psi_X \in \mathfrak{so}(E)$ by the adjointness of K_X and K_X^*, the connection $\nabla^\oplus$ is metric, and its blocks are exactly the stated ones. If $\tilde{\nabla}$ is another metric connection with the same diagonal blocks and the same $TM \to N$ block, then $\tilde{\nabla} - (\nabla^h \oplus \nabla^\perp)$ has the form $\left(\begin{smallmatrix} 0 & B_X \\ K_X & 0 \end{smallmatrix}\right)$, and metricity forces $B_X = -K_X^*$. Hence $\tilde{\nabla} = \nabla^\oplus$. $\square$

B Curvature Channels

Once the assembled connection is fixed, its curvature decomposes according to the same split geometry.

Proposition II.3 (Curvature Decomposition)**.** *The curvature $\mathcal{F}^{\oplus}$ of $\nabla^{\oplus}$ decomposes as*

$$\mathcal{F}^{\oplus} = \begin{pmatrix} \mathcal{G} & -\mathcal{C}^* \\ \mathcal{C} & \mathcal{S} \end{pmatrix} \qquad \textit{where} \qquad \mathcal{G} = R^h - (K^* \wedge K), \quad \mathcal{C} = D^{\mathrm{Hom}} K, \quad \mathcal{S} = R^{\perp} - (K \wedge K^*).$$

Proof. Write $\nabla^{\oplus} = \nabla^{\mathrm{prod}} + \Psi$. Its curvature is

$$\mathcal{F}^{\oplus} = F^{\nabla^{\mathrm{prod}}} + D^{\nabla^{\mathrm{prod}}}\Psi + \Psi \wedge \Psi, \qquad F^{\nabla^{\mathrm{prod}}} = \begin{pmatrix} R^h & 0 \\ 0 & R^{\perp} \end{pmatrix}.$$

Since Ψ is off-diagonal, $D^{\nabla^{\mathrm{prod}}}\Psi$ is also off-diagonal, with mixed block $D^{\mathrm{Hom}}K$ and adjoint block $-(D^{\mathrm{Hom}}K)^*$. The graded wedge formulas above give

$$\Psi \wedge \Psi = \begin{pmatrix} -(K^* \wedge K) & 0 \\ 0 & -(K \wedge K^*) \end{pmatrix}.$$

Adding the three contributions yields the stated decomposition. □

Theorem II.4 (Integrability Equivalence)**.** *For GCR data $(h, K, \nabla^{\perp})$, $\mathcal{F}^{\oplus} = 0$ if and only if $(\mathcal{G}, \mathcal{C}, \mathcal{S}) = 0$. Equivalently, flatness of $\nabla^{\oplus}$ is exactly the Gauss–Codazzi–Ricci system.*

Proof. By Proposition II.3, $\mathcal{F}^{\oplus} = 0$ if and only if the three channels $\mathcal{G}$, $\mathcal{C}$, and $\mathcal{S}$ vanish. These are exactly the Gauss, Codazzi, and Ricci equations written in block form for the assembled connection; see [3, Vol. II, Ch. VII]. □

C Internal Gauge Equivariance

The assembled orthogonal geometry is natural under internal gauge on the normal bundle.

Corollary II.5 (Internal Gauge Equivariance)**.** *The assembled connection $\nabla^{\oplus}$ and its curvature $\mathcal{F}^{\oplus}$ are $\mathcal{G}_N$-equivariant. In particular, for every $\gamma \in \mathcal{G}_N$,*

$$\mathcal{F}^{\oplus} \mapsto \gamma^{-1}\mathcal{F}^{\oplus}\gamma, \tag{II.3}$$

so the block decomposition $\mathcal{F}^{\oplus} = (\mathcal{G}, -\mathcal{C}^*; \mathcal{C}, \mathcal{S})$ is preserved.

Proof. Internal normal gauge acts trivially on TM and orthogonally on N, hence by conjugation on $E = TM \oplus N$. Therefore $\nabla^{\oplus} \mapsto \gamma^{-1}\nabla^{\oplus}\gamma$, and the usual covariance of curvature gives (II.3). Since the splitting $E = TM \oplus N$ is preserved by the gauge action, so is the block decomposition of $\mathcal{F}^{\oplus}$. □

D Canonical Pairing and Bianchi Identities

The assembled curvature carries two further structural consequences, given by the canonical pairing and the Bianchi identity.

Proposition II.6 (Block Energy Splitting)**.** *The canonical quadratic pairing satisfies*

$$\|\mathcal{F}^{\oplus}\|^2 = \|\mathcal{G}\|^2 + 2\|\mathcal{C}\|^2 + \|\mathcal{S}\|^2.$$

Proof. The splitting $\mathfrak{so}(E) = \mathfrak{so}(TM) \oplus \big(\mathrm{Hom}(TM,N) \oplus \mathrm{Hom}(N,TM)\big) \oplus \mathfrak{so}(N)$ is orthogonal under $-\mathrm{tr}$, so cross terms vanish. The off-diagonal contribution occurs twice through the $TM \to N$ and $N \to TM$ adjoint slots, yielding the coefficient 2. □

For the Bianchi decomposition, write the graded commutator as

$$[\alpha,\beta]_{\mathrm{gr}} := \alpha \wedge \beta - (-1)^{|\alpha||\beta|}\beta \wedge \alpha. \tag{II.4}$$

Let

$$D^{\mathrm{prod}} := D^{\nabla^{\mathrm{prod}}}$$

denote the exterior covariant derivative on $\Omega^*(M,\mathfrak{so}(E))$ associated with the product connection $\nabla^{\mathrm{prod}} = \nabla^h \oplus \nabla^\perp$. Then

$$D^\oplus = D^{\mathrm{prod}} + [\Psi,\cdot]_{\mathrm{gr}},$$

Theorem II.7 (Coupled Bianchi System)**.** *The identity $D^\oplus \mathcal{F}^\oplus = 0$ decomposes as*

$$D^T\mathcal{G} + [K^*,\mathcal{C}] = 0, \qquad D^{\mathrm{Hom}}\mathcal{C} + K \wedge \mathcal{S} - \mathcal{G} \wedge K = 0, \qquad D^\perp \mathcal{S} + [K,\mathcal{C}^*] = 0.$$

Proof. By the second Bianchi identity, $D^\oplus \mathcal{F}^\oplus = 0$. Using $\mathcal{F}^\oplus = \left(\begin{smallmatrix} \mathcal{G} & -\mathcal{C}^* \\ \mathcal{C} & \mathcal{S} \end{smallmatrix}\right)$ and $D^\oplus = D^{\mathrm{prod}} + [\Psi,\cdot]_{\mathrm{gr}}$, projection onto the (TT), (TN), and (NN) blocks gives the three displayed identities. □

III Admissible Connection Space

Once the orthogonal geometry is assembled, one may consider the full space of such split metric connections. Admissibility is then expressed by flatness within that space.
Define the admissible connection space by

$$\mathcal{A}_\oplus := \{\nabla^\oplus = \nabla^h \oplus \nabla^\perp + \Psi(K) : \nabla^\perp \in \Omega^1(M,\mathfrak{so}(N)),\ K \in \Gamma(\mathrm{Sym}^2 T^*M \otimes N)\}.$$

Definition III.1 (Admissible Flatness System)**.** *Internal normal gauge is $\mathcal{G}_N := \Gamma(\mathrm{SO}(N))$. The flatness operator is $\Phi : \mathcal{A}_\oplus \to \Omega^2(M,\mathfrak{so}(E))$, $\Phi(\nabla^\oplus) := \mathcal{F}^\oplus$. The flat locus is $\mathcal{Z}_{\mathrm{flat}} := \Phi^{-1}(0) \subset \mathcal{A}_\oplus$.*

Definition III.2 (Reduced Tangent and Reduced Symbol)**.** *At $\nabla^\oplus \in \mathcal{A}_\oplus$, let*

$$\mathbf{G} := D(\mathrm{gauge})\big|_{\nabla^\oplus} : \Gamma(\mathfrak{so}(N)) \to T_{\nabla^\oplus}\mathcal{A}_\oplus$$

denote the infinitesimal internal normal gauge map. The reduced tangent *is*

$$T^{\mathrm{red}}_{\nabla^\oplus}\mathcal{A}_\oplus \ := \ T_{\nabla^\oplus}\mathcal{A}_\oplus \Big/ \mathrm{im}\,(\mathbf{G}),$$

and at $(x,\xi) \in T^*M \setminus \{0\}$ the reduced symbol domain *is the quotient*

$$\sigma^{\mathrm{red}}_\xi(\eta) \ := \ \sigma_\xi(\eta) \Big/ \mathrm{im}\,\big(\sigma_\xi(\mathbf{G})\big).$$

At flat compatible backgrounds with $K_0 = 0$, the mixed slot survives gauge reduction because the internal gauge action is order zero there and symbol-invisible in that slot; see Lemmas IV.0.3 and IV.3.1.

Proposition III.3 (Kinematic Constraint Geometry)**.** *The triple $(\mathcal{A}_\oplus, \Phi, \mathcal{G}_N)$ has the following structure: $\mathcal{G}_N$ preserves $\mathcal{A}_\oplus$, the curvature map is gauge-equivariant, and for $\eta \in T_{\nabla^\oplus}\mathcal{A}_\oplus$,*

$$D\Phi|_{\nabla^\oplus}(\eta) = D^{\nabla^\oplus}\eta.$$

Proof. Internal normal gauge acts by block-diagonal orthogonal automorphisms of $E = TM \oplus N$, hence preserves metricity, the fixed tangent block, and the symmetry of K; therefore it preserves $\mathcal{A}_{\oplus}$. Curvature is gauge-covariant, so

$$\Phi(g^{-1} \cdot \nabla) = g^{-1}\Phi(\nabla)g.$$

For a smooth variation $\nabla_t = \nabla^{\oplus} + t\eta$, the standard curvature variation formula gives

$$\frac{d}{dt}\Big|_{t=0} F^{\nabla_t} = D^{\nabla^{\oplus}}\eta.$$

Hence $D\Phi|_{\nabla^{\oplus}}(\eta) = D^{\nabla^{\oplus}}\eta$. □

IV Mixed Sector

The mixed channel is the place where the residual compatibility geometry remains open. The strength of the flatness condition on that channel is measured by the reduced principal symbol of the linearized curvature map. At flat compatible backgrounds with vanishing mixed basepoint K_0, the mixed slot survives internal gauge reduction and carries the non-elliptic directions that govern the rest of the analysis.

A Reduced Symbol

The reduced symbol is formulated at a flat compatible background together with the slotwise behavior of the internal gauge action. The underlying symbol identities are standard, but in the present split setting they show that the mixed slot remains visible after reduction.

Lemma IV.0.1 (Symbol of covariant exterior derivative)**.** *Let ∇ be a connection on a vector bundle $V \to M$. For $\alpha \in \Omega^p(M, V)$ and $\xi \in T_x^*M \setminus \{0\}$,*

$$\sigma_\xi(D^\nabla)(\alpha) = \xi \wedge \alpha.$$

The graded commutator on $\Omega^*(M, \mathfrak{so}(E))$ is

$$[\alpha, \beta]_{\mathrm{gr}} := \alpha \wedge \beta - (-1)^{|\alpha||\beta|}\beta \wedge \alpha \quad \textit{for homogeneous } \alpha, \beta. \tag{IV.1}$$

Proof. In a local trivialization, $D^\nabla = d + A$ with A order zero, so the principal symbol is that of d, namely wedge with ξ. □

Lemma IV.0.2 (Linearization of curvature and its symbol)**.** *For a connection ∇ on V, the curvature map $\nabla \mapsto F^\nabla$ has linearization*

$$DF^\nabla(\eta) = D^\nabla\eta, \qquad \eta \in \Omega^1(M, \mathrm{End}(V)).$$

Hence

$$\sigma_\xi(DF^\nabla)(\eta) = \xi \wedge \eta.$$

Proof. The first identity is the standard curvature variation formula. The symbol identity follows from Lemma IV.0.1. □

At a flat compatible background, the remaining point is the slotwise behavior of the internal gauge action, especially on the mixed component.

Lemma IV.0.3 (Internal Gauge Action by Slots)**.** *In the block decomposition $E = TM \oplus N$ and internal gauge $\phi \in \Gamma(\mathfrak{so}(N))$, the infinitesimal action satisfies*

$$\delta_\phi(\nabla^\perp) = D^\perp \phi, \qquad \delta_\phi K = -\,\phi \cdot K,$$

where $\phi \cdot K$ denotes the induced tensor representation of $\mathfrak{so}(N)$ on $\mathrm{Sym}^2 T^*M \otimes N$ via the standard action of $\mathrm{SO}(N)$ on N (in adapted orthonormal components: $(\phi \cdot K)_{ij}^{\alpha} = \sum_\beta \phi^\alpha{}_\beta K_{ij}^\beta$). Thus the action is order one on the normal-connection slot and order zero on the mixed slot; the mixed component is algebraic and linear in the background K.

Proof. Differentiate the gauge action $\nabla^\perp \mapsto g^{-1}\nabla^\perp g$ and $K \mapsto g^{-1} \cdot K$ at $g = \exp(t\phi)$. □

Definition IV.1 (Flat point and frozen flat chart)**.** *A* flat point *is a connection $\nabla^\oplus \in \mathcal{Z}_{\mathrm{flat}}$, equivalently $F^{\nabla^\oplus} = 0$. A* frozen flat chart *at a point x is a choice of local coordinates and normal gauge such that the Levi-Civita coefficients of h and the normal-connection coefficients of $\nabla^\perp$ vanish at x. On a simply connected open set U,* frozen flat on U *means these coefficients vanish identically on U.*

Theorem IV.2 (Principal symbol of the curvature linearization)**.** *For any background $\nabla^\oplus \in \mathcal{A}_\oplus$ and $\eta \in T_{\nabla^\oplus}\mathcal{A}_\oplus$,*

$$\sigma_\xi(\mathbf{L}_{\mathrm{full}})(\eta) = \xi \wedge \eta.$$

Proof. This is Lemma IV.0.2 with $V = E$. □

Corollary IV.3 (Block projection of the principal symbol)**.** *Let $\eta = \iota(\delta\nabla^\perp, \delta K)$. Then*

$$\mathrm{pr}_{TT,TN,NN}\big(D\Phi|_{\nabla^\oplus}(\eta)\big) = D(\mathcal{G},\mathcal{C},\mathcal{S})|_{(h,K,\nabla^\perp)}(\delta\nabla^\perp, \delta K),$$

and

$$\mathrm{pr}_{TT,TN,NN}\big(\sigma_\xi(\mathbf{L}_{\mathrm{full}})(\eta)\big) = \mathrm{pr}_{TT,TN,NN}(\xi \wedge \eta).$$

Proof. The first identity is the block definition of $(h, K, \nabla^\perp) \mapsto (\mathcal{G},\mathcal{C},\mathcal{S})$, and the second is Theorem IV.2 projected to the three blocks. □

Lemma IV.3.1 (Mixed-slot gauge linearization at $K_0 = 0$)**.** *For internal gauge parameter $\phi \in \Gamma(\mathfrak{so}(N))$ and background $(h, K_0, \nabla^\perp)$,*

$$\delta_\phi K = -\,\phi \cdot K_0.$$

Hence at backgrounds with $K_0 = 0$, the full linearized gauge action on the mixed slot vanishes:

$$\mathrm{pr}_{\delta K}(\mathbf{G}\,\phi) = 0, \qquad \mathrm{pr}_{\delta K}\big(\sigma_\xi(\mathbf{G})\,\phi\big) = 0.$$

Proof. The first identity is Lemma IV.0.3. It is algebraic in ϕ and linear in K_0, so both the full mixed component and its principal symbol vanish at $K_0 = 0$. □

At a flat compatible background with $K_0 = 0$ and $\delta\nabla^\perp = 0$, the mixed block reduces to the operator

$$L := D^{\mathrm{Hom}}$$

on the δK-slot, and its principal symbol is

$$(\sigma_\xi L)(\delta K)_{ij}{}^{\alpha}{}_k = \xi_i\, \delta K_{jk}^\alpha - \xi_j\, \delta K_{ik}^\alpha.$$

Hence $\sigma_\xi(L)(\delta K) = 0$ is equivalent to $\xi \wedge \delta K = 0$.

Theorem IV.4 (Reduced Symbol Kernel)**.** *Let $n \geq 2$. For every nonzero $\xi \in T_x^*M$,*

$$\sigma_\xi(L) : \mathrm{Sym}^2(T_x^*M) \otimes N_x \longrightarrow \Lambda^2(T_x^*M) \otimes T_x^*M \otimes N_x$$

has

$$\ker \sigma_\xi(L) = \{\xi \odot \xi \otimes \nu : \nu \in N_x\}, \qquad \dim \ker \sigma_\xi(L) = k.$$

At $K_0 = 0$, internal gauge is symbol-invisible in the mixed slot; these kernel directions survive gauge reduction.

Proof. The symbol equation is $\xi \wedge \delta K = 0$. In components,

$$(\sigma_\xi L)(\delta K)_{ij}{}^{\alpha}{}_k = \xi_i\, \delta K^{\alpha}_{jk} - \xi_j\, \delta K^{\alpha}_{ik} = 0.$$

For $\xi \neq 0$, antisymmetry in (i, j) forces δK to lie in the symmetric line generated by $\xi \odot \xi$ in the cotangent variables; thus the kernel of

$$\xi \wedge \cdot : \mathrm{Sym}^2(T_x^*M) \otimes N_x \to \Lambda^2(T_x^*M) \otimes T_x^*M \otimes N_x$$

consists exactly of tensors of the form $\xi \odot \xi \otimes \nu$ with $\nu \in N_x$. Hence

$$\ker \sigma_\xi(L) = \{\xi \odot \xi \otimes \nu : \nu \in N_x\}, \qquad \dim \ker \sigma_\xi(L) = k.$$

By Lemma IV.3.1, $\mathrm{pr}_{\delta K}(\sigma_\xi(\mathbf{G})\phi) = 0$ at $K_0 = 0$, so these kernel directions survive principal gauge reduction. □

B Non-Ellipticity

The reduced symbol kernel already forces non-injectivity at the level of the gauge-reduced compatibility operator.
The corresponding smooth equivariant bundle packaging over S^*M is recorded in Appendix B.

Theorem IV.5 (Reduced Non-Ellipticity)**.** *In the background-fixed internal-gauge setting with $\mathcal{G}_N = \Gamma(\mathrm{SO}(N))$, at each flat compatible background $\nabla^\oplus \in \mathcal{Z}_{\mathrm{flat}}$ with $K_0 = 0$ and for every nonzero covector ξ, the reduced principal symbol of $(\mathcal{A}_\oplus, \Phi, \mathcal{G}_N)$ fails injectivity for $n \geq 2$.*

Proof. Fix $0 \neq \nu \in N_x$ and set $\delta K := \xi \odot \xi \otimes \nu$. By Theorem IV.4, $\delta K \in \ker \sigma_\xi(L)$. Embed the K-sector element as

$$\eta := \iota(0, \delta K) \in (T_{\nabla^\oplus}\mathcal{A}_\oplus)_x.$$

By Corollary IV.3, the mixed block of $\sigma_\xi(\mathbf{L}_{\mathrm{full}})(\eta)$ is $\sigma_\xi(L)(\delta K) = 0$, while the Gauss and Ricci blocks vanish because η has no $\delta\nabla^\perp$ component. Hence $\sigma_\xi(\mathbf{L}_{\mathrm{full}})(\eta) = 0$. By Lemma IV.3.1, $\mathrm{pr}_{\delta K}(\sigma_\xi(\mathbf{G})\phi) = 0$ at $K_0 = 0$ for every ϕ, so the class of η is nonzero in the reduced domain. Therefore the reduced symbol is non-injective. □

Definition IV.6 (Admissible internal gauge fixing at flat compatible backgrounds)**.** *An admissible internal gauge fixing in the background-fixed internal-gauge setting is a first-order operator*

$$\mathcal{Q} : T_{\nabla^\oplus}\mathcal{A}_\oplus \to \Gamma(W)$$

whose principal symbol depends only on the normal-connection slot and vanishes on the mixed slot at backgrounds with $K_0 = 0$.

Theorem IV.7 (No-Restoration Theorem)**.** *Let $\nabla^{\oplus} \in \mathcal{Z}_{\mathrm{flat}}$ be a flat compatible background with $K_0 = 0$, and let $n \geq 2$. For any admissible internal gauge fixing $\mathcal{Q}$, the augmented reduced principal symbol is non-injective for every $\xi \neq 0$.*

Proof. By Theorem IV.5, the reduced kernel contains a nonzero class represented by a mixed-direction element $\eta = \iota(0, \delta K)$. By admissibility, $\sigma_\xi(\mathcal{Q})(\eta) = 0$ at $K_0 = 0$, so the same class lies in the kernel of the reduced augmented symbol. Hence injectivity cannot be restored. □

On a closed manifold, this symbol obstruction has the following analytic consequence.

Proposition IV.8 (Frozen Codazzi non-elliptic)**.** *At the formal level, the map $L = D^{\mathrm{Hom}} : \Gamma(\mathrm{Sym}^2 T^*M \otimes N) \to \Gamma(\Lambda^2 T^*M \otimes T^*M \otimes N)$ has non-injective symbol and is therefore non-elliptic.*

Proof. Theorem IV.4 gives $\dim \ker \sigma_\xi(L) = k$ for every nonzero ξ. Therefore $\sigma_\xi(L)$ is never injective for $\xi \neq 0$, so ellipticity fails. □

Theorem IV.9 (Non-Fredholm Codazzi Linearization)**.** *Let M^n be compact, smooth, and without boundary, with $n \geq 2$. For every $s \in \mathbb{R}$, the frozen Codazzi operator*

$$L : H^s(\mathrm{Sym}^2 T^*M \otimes N) \to H^{s-1}(\Lambda^2 T^*M \otimes T^*M \otimes N)$$

is not Fredholm for $n \geq 2$.

Proof. Fix $(x, \xi) \in S^*M$ and choose $0 \neq a \in \ker \sigma_\xi(L)$, which exists by Theorem IV.4. In local coordinates, take

$$u_\lambda(y) = e^{i\lambda\varphi(y)} a(y), \qquad d\varphi_x = \xi.$$

Then the principal term vanishes, and standard high-frequency estimates give

$$\|Lu_\lambda\|_{H^{s-1}} = o(1)\|u_\lambda\|_{H^s} \qquad (\lambda \to \infty).$$

After normalization $\|u_\lambda\|_{H^s} = 1$, this yields a high-frequency approximate null sequence. Since L is a first-order differential operator on the closed manifold M, Fredholmness is equivalent to ellipticity for the associated classical pseudodifferential operator [5, 4, 6, Vol. III, Ch. 20]. By Proposition IV.8, ellipticity fails. Therefore L is not Fredholm. □

V Mixed-Sector Structure

At the symbol level, the mixed sector carries a residual kernel. On flat trivialized domains, that kernel is generated by Hessian potentials, and this local generation passes directly to the reduced formal tangent. This section records those two structural consequences.

A Hessian Generation

On a flat trivialized domain, the mixed kernel has a local Hessian description. The kernel of D^{Hom} on symmetric mixed tensors is exactly the Hessian image, so the mixed directions are locally generated by second derivatives.

Theorem V.1 (Hessian Generation of the Mixed Kernel)**.** *On a connected simply connected open set U with flat trivialized induced connection on $T^*U \otimes N$, the kernel*

$$\ker(D^{\mathrm{Hom}}) \cap \Gamma(\mathrm{Sym}^2 T^*U \otimes N)$$

is given by

$$\mathrm{Im}\big(\mathrm{Hess} : C^\infty(U, N) \to \Gamma(\mathrm{Sym}^2 T^*U \otimes N)\big).$$

Proof. In a frozen flat chart, the induced connection on $T^*U \otimes N$ is $d \otimes \mathrm{id}_N$, so $D^{\mathrm{Hom}}\delta K = 0$ if and only if for each fixed index pair (k, α) the 1-form $\sum_i \delta K^\alpha_{ik}\, dx^i$ is closed:

$$\partial_i \delta K^\alpha_{jk} - \partial_j \delta K^\alpha_{ik} = 0.$$

Since U is simply connected, the Poincaré lemma gives

$$\delta K^\alpha_{ik} = \partial_i V^\alpha_k$$

for some V^α_k. Symmetry $\delta K^\alpha_{ik} = \delta K^\alpha_{ki}$ yields the integrability condition

$$\partial_i V^\alpha_k = \partial_k V^\alpha_i.$$

A second Poincaré step gives u^α with $V^\alpha_i = \partial_i u^\alpha$, hence

$$\delta K^\alpha_{ik} = \partial_i \partial_k u^\alpha.$$

Conversely, $\mathrm{Hess}(u) \in \ker D^{\mathrm{Hom}}$ because mixed partial derivatives commute. □

Proposition V.2 (Local Exactness of the Mixed Hessian Sequence)**.** *Let U be a connected simply connected open set with flat trivialized induced connection on $T^*U \otimes N$. The sequence*

$$C^\infty(U, N) \xrightarrow{\mathrm{Hess}} \Gamma(\mathrm{Sym}^2 T^*U \otimes N) \xrightarrow{D^{\mathrm{Hom}}} \Gamma(\Lambda^2 T^*U \otimes T^*M \otimes N)$$

is exact at $\Gamma(\mathrm{Sym}^2 T^*U \otimes N)$.

Proof. This is exactly Theorem V.1. □

B Reduced Formal Tangent

The Hessian-generated mixed directions pass directly to the reduced formal tangent.

Theorem V.3 (Infinite-Dimensional Mixed Reduced Tangent)**.** *Let $n \geq 2$ and $\nabla^\oplus \in \mathcal{Z}_{\mathrm{flat}}$ be formal. On each connected simply connected open set U with flat trivialized induced connection on $T^*U \otimes N$, the reduced tangent quotient $T^{\mathrm{red}}_{\nabla^\oplus}\mathcal{Z}_{\mathrm{flat}}$ contains an infinite-dimensional mixed-sector subspace.*

Proof. Let U be as stated, and impose $K_0 = 0$ on U. For $u \in C^\infty(U, N)$, set $\delta K = \mathrm{Hess}(u)$. By Theorem V.1,

$$D^{\mathrm{Hom}}(\delta K) = 0.$$

Hence $\eta = \iota(0, \delta K) \in \ker(D\Phi|_{\nabla^\oplus})$. By Lemma IV.3.1, internal gauge has zero mixed-slot projection at $K_0 = 0$, so the assignment

$$\mathrm{Im}(\mathrm{Hess}) \longrightarrow T^{\mathrm{red}}_{\nabla^\oplus}\mathcal{Z}_{\mathrm{flat}}, \qquad \delta K \longmapsto [\,\iota(0, \delta K)\,]$$

is injective. In a frozen flat trivialization, choose a nonzero constant normal vector $\nu \in \Gamma(U, N)$. Then the Hessians of the sections $u_m := x_1^m \nu$ for $m \geq 2$ are linearly independent. Hence $\mathrm{Im}(\mathrm{Hess})$ is infinite-dimensional, and so $T^{\mathrm{red}}_{\nabla^\oplus}\mathcal{Z}_{\mathrm{flat}}$ contains an infinite-dimensional mixed-sector subspace. □

VI Codimension-One

In codimension one the general mixed compatibility geometry undergoes a structural collapse. The Ricci channel disappears, leaving only the mixed and Gauss channels. This section shows that the surviving codimension-one remnant is closed in the natural linear metric vector-scalar category and uniquely determines the fixed-metric vacuum Einstein constraint structure.

A Rank-One Channel Reduction

Theorem VI.1 (Rank-One Channel Reduction)**.** *Assume* $\operatorname{rank} N = 1$*. Then* $\mathfrak{so}(N) = 0$*, the normal connection is flat, and the Ricci channel vanishes identically:*

$$\mathcal{S} \equiv 0.$$

Accordingly,

$$\mathcal{F}^{\oplus} = \begin{pmatrix} \mathcal{G} & -\mathcal{C}^* \\ \mathcal{C} & 0 \end{pmatrix},$$

so the only surviving codimension-one channels are the mixed channel $\mathcal{C}$ *and the Gauss channel* $\mathcal{G}$*.*

Proof. In rank one, $\mathfrak{so}(N) = 0$, so a metric connection on N is necessarily flat. By Proposition II.3, the Ricci block $\mathcal{S}$ is $\mathfrak{so}(N)$-valued, hence vanishes identically. The displayed block form follows immediately. □

B Intrinsic Constraint Map

Definition VI.2 (Intrinsic Constraint Map)**.** *Let* $\mathcal{A}_{\oplus}^{(1)} \subset \mathcal{A}_{\oplus}$ *be the rank-one slice. Define the contractions*

$$\mathrm{Contr}(\mathcal{C})(X) := \sum_j \mathcal{C}(e_j, X)e_j, \qquad \mathrm{tr}_h(\mathcal{G}) := \sum_{i,j} h(\mathcal{G}(e_i, e_j)e_j, e_i).$$

The intrinsic constraint map *is*

$$\mathcal{K}_{\mathrm{comp}} : \mathcal{A}_{\oplus}^{(1)} \to \Gamma(T^*M) \oplus C^\infty(M), \qquad \mathcal{K}_{\mathrm{comp}}(\nabla^{\oplus}) := (\mathrm{Contr}(\mathcal{C}), \mathrm{tr}_h(\mathcal{G})).$$

Its zero locus is $\mathcal{Z}_{\mathrm{comp}}^{(1)} := \mathcal{K}_{\mathrm{comp}}^{-1}(0)$*.*

Theorem VI.3 (Intrinsic Constraint Sector)**.** $\mathcal{K}_{\mathrm{comp}}$ *is intrinsically defined on* $\mathcal{A}_{\oplus}^{(1)}$ *and natural under orthogonal line-bundle isometries. Its zero locus defines the intrinsic constraint sector in codimension one.*

Proof. The contraction formulas in Definition VI.2 use only the metric h, the orthogonal splitting $E = TM \oplus N$, and the rank-one curvature blocks $(\mathcal{C}, \mathcal{G})$. They are therefore independent of the chosen local h-orthonormal frame and invariant under orthogonal transformations of the fibers. Under $n \mapsto -n$ in a unit-normal trivialization, only the scalar representative of the intrinsic N-valued form $\mathrm{Contr}(\mathcal{C})$ changes sign. Hence $\mathcal{K}_{\mathrm{comp}}$ is intrinsic and its zero locus is well defined. □

Definition VI.4 (Vacuum Constraint Map)**.** *After unit-normal trivialization* $N \cong M \times \mathbb{R}$*, define the vacuum constraint map by*

$$\mathcal{K}_{\mathrm{vac}}^{(h)}(K) := \big(\nabla^j K_{ij} - \nabla_i \mathrm{tr}_h K,\ \mathrm{scal}_h - |K|_h^2 + (\mathrm{tr}_h K)^2\big).$$

C Vacuum Constraint Identification

Theorem VI.5 (Codimension-One Constraint Identification)**.** *Assume* $\operatorname{rank} N = 1$ *and choose a unit-normal trivialization. Then*

$$\mathcal{K}_{\mathrm{comp}} = \mathcal{K}_{\mathrm{vac}}^{(h)}.$$

Proof. By Proposition II.3, the mixed block is $\mathcal{C} = D^{\mathrm{Hom}}K$. In a unit normal trivialization, K is an ordinary symmetric 2-tensor and

$$\mathcal{C}_{ijk} = \nabla_i K_{jk} - \nabla_j K_{ik}.$$

Contracting the second and third tangent slots gives

$$(\mathrm{Contr}(\mathcal{C}))_i = h^{jk}\mathcal{C}_{jik} = \nabla^j K_{ij} - \nabla_i \mathrm{tr}_h K.$$

For the Gauss block,

$$\mathcal{G} = R^h - (K^* \wedge K).$$

Scalar contraction yields

$$\mathrm{tr}_h(\mathcal{G}) = \mathrm{scal}_h - \mathrm{tr}_h(K^* \wedge K) = \mathrm{scal}_h - |K|_h^2 + (\mathrm{tr}_h K)^2.$$

Here $\mathrm{tr}_h(K^* \wedge K) = |K|_h^2 - (\mathrm{tr}_h K)^2$ by the symmetry of K in rank one. Therefore the two components agree with Definition VI.4; these are exactly the standard fixed-metric vacuum Einstein constraint expressions [7, 8, 9]. □

D Classification

In codimension one, the surviving mixed and Gauss channels admit only one natural vector output and one natural scalar output.

Definition VI.6 (Natural Vector-Scalar Contractions)**.** *Let $n \geq 2$ and let $V = T_x M$ be equipped with its metric h. Let*

$$\mathcal{W} := \Lambda^2 V^* \otimes N$$

denote the mixed channel bundle, and let $\mathcal{R}$ denote the bundle of algebraic curvature tensors on V. A natural vector-scalar contraction *is an $O(V)$-equivariant linear map*

$$\mathcal{W} \to V^* \qquad \text{or} \qquad \mathcal{R} \to \mathbb{R}.$$

Theorem VI.7 (Mixed-Channel Classification)**.** *Let $n \geq 2$. The space $\mathrm{Hom}_{O(V)}(\mathcal{W}, V^*)$ is one-dimensional, generated by $C \mapsto h^{jk}C_{ijk}$. Every natural linear metric vector contraction of the codimension-one mixed channel is a scalar multiple of $\mathrm{Contr}(\mathcal{C})$.*

Proof. Standard invariant theory [1] reduces the problem to metric contractions of a tensor in $\mathcal{W}$. Up to relabeling, the only possibilities are

$$A_i := h^{jk}C_{ijk}, \qquad B_i := h^{jk}C_{jik}, \qquad D_i := h^{jk}C_{jki}.$$

Antisymmetry gives $B_i = -A_i$. The cyclic identity gives $D_i = A_i$. Hence every contraction is a scalar multiple of $A_i = h^{jk}C_{ijk}$, so $\mathrm{Hom}_{O(V)}(\mathcal{W}, V^*)$ is one-dimensional and generated by $\mathrm{Contr}(\mathcal{C})$. □

Theorem VI.8 (Gauss-Channel Classification)**.** *Let $n \geq 2$. The space $\mathrm{Hom}_{O(V)}(\mathcal{R}, \mathbb{R})$ is one-dimensional, generated by scalar-curvature contraction. Every natural linear metric scalar contraction of the codimension-one Gauss channel is a scalar multiple of $\mathrm{tr}_h(\mathcal{G})$.*

Proof. Standard invariant theory [1] reduces the problem to full metric contractions of a tensor in $\mathcal{R}$. Up to relabeling, the only possibilities are

$$h^{ik}h^{jl}R_{ijkl}, \qquad h^{ij}h^{kl}R_{ijkl}, \qquad h^{il}h^{jk}R_{ijkl}.$$

The middle one vanishes by antisymmetry in (i, j), and the last equals the first up to sign by antisymmetry in (k, l). Hence every natural linear metric scalar contraction is a scalar multiple of $h^{ik}h^{jl}R_{ijkl} = \mathrm{tr}_h(\mathcal{G})$. □

E Exclusion

The remaining alternatives are excluded by parity and invariant type.

Lemma VI.8.1 (No Mixed-to-Scalar Output)**.** $\mathrm{Hom}_{O(V)}(\mathcal{W}, \mathbb{R}) = 0$ *for* $n \geq 2$*. There is no natural linear metric scalar output from the mixed channel.*

Proof. Every tensor in $\mathcal{W}$ has odd covariant degree. A scalar contraction built only from the metric would have to pair off all covariant indices, which is impossible. Hence the hom-space vanishes. □

Lemma VI.8.2 (No Gauss-to-Vector Output)**.** $\mathrm{Hom}_{O(V)}(\mathcal{R}, V^*) = 0$ *for* $n \geq 2$*. There is no natural linear metric vector output from the Gauss channel.*

Proof. A vector output would require leaving exactly one covariant index unpaired after contracting one algebraic curvature tensor with the metric. This is impossible: metric contractions remove indices in pairs, so the result has either rank 0 or even positive rank. Therefore no natural linear metric map $\mathcal{R} \to V^*$ exists. □

F Closure

The classification and exclusion results together close the codimension-one remnant.

Definition VI.9 (Closed Admissibility Remnant)**.** *Assume* $\operatorname{rank} N = 1$*. The* codimension-one compatibility remnant *is the surviving mixed/Gauss pair* $(\mathcal{C}, \mathcal{G})$*. It is called* closed *in the natural linear metric vector-scalar category if:*

1. *every natural linear metric vector-scalar output of the codimension-one compatibility geometry factors through the pair* $(\mathrm{Contr}(\mathcal{C}), \mathrm{tr}_h(\mathcal{G}))$*;*
2. *there are no mixed-to-scalar outputs and no Gauss-to-vector outputs in the natural linear metric category.*

Theorem VI.10 (Codimension-One Closure)**.** *Assume* $\operatorname{rank} N = 1$*. Then the mixed/Gauss residual is closed: it exhausts all natural linear metric vector-scalar outputs of codimension-one compatibility geometry, and no cross-channel outputs exist.*

Proof. By Theorem VI.1, rank one removes the Ricci channel. By Theorems VI.7 and VI.8, the surviving mixed and Gauss channels each admit exactly one natural linear metric output type, up to scale. By Lemmas VI.8.1 and VI.8.2, no cross-channel outputs exist. Hence the codimension-one residual is closed in the sense of Definition VI.9. □

G Determination

Theorem VI.11 (Codimension-One Determination)**.** *Assume* $\operatorname{rank} N = 1$*. In the natural linear metric vector-scalar category, every codimension-one output is of the form*

$$\Theta = \big(a\ \mathrm{Contr}(\mathcal{C}),\ b\,\mathrm{tr}_h(\mathcal{G})\big)$$

After unit-normal identification, the resulting admissibility structure is exactly the fixed-metric vacuum Einstein constraint structure.

Proof. By Theorem VI.10, every admissible output is built from the mixed/Gauss residual. By Theorems VI.7 and VI.8, its vector and scalar components are therefore scalar multiples of $\mathrm{Contr}(\mathcal{C})$ and $\mathrm{tr}_h(\mathcal{G})$. This determines the codimension-one admissibility structure, and the final identification is Theorem VI.5. □

This determination is internal to the frozen-tangential compatibility geometry. No ambient Einstein equation is used.

H Rigidity

Corollary VI.12 (Codimension-One Rigidity)**.** *Assume* $\operatorname{rank} N = 1$*, and let* Θ *be a nontrivial natural linear metric codimension-one vector-scalar output. Then*

$$\Theta^{-1}(0) = \mathcal{K}_{\mathrm{comp}}^{-1}(0).$$

After unit-normal identification,

$$\Theta^{-1}(0) = \{K : \mathcal{K}_{\mathrm{vac}}^{(h)}(K) = 0\}.$$

Proof. By Theorem VI.11, $\Theta = (a\ \mathrm{Contr}(\mathcal{C}),\ b\,\mathrm{tr}_h(\mathcal{G}))$ with $(a, b) \neq (0, 0)$. Hence $\Theta = 0$ exactly when $\mathrm{Contr}(\mathcal{C}) = 0$ and $\mathrm{tr}_h(\mathcal{G}) = 0$, so

$$\Theta^{-1}(0) = \mathcal{K}_{\mathrm{comp}}^{-1}(0).$$

The second equality is Theorem VI.5. □

VII Concluding Remark

The structure of admissibility implies a larger argument. Rather than beginning with spacetime and deriving its constraints, one may begin with structural admissibility and study how the geometry determines its own constraints. In codimension one, this reaches the point at which the spacetime metric structure emerges. All of this takes place without imposing variation or dynamics from the intrinsic metric of spacetime.

In future work, the admissible representation, variational, differential, and symbol structures close into a single structural equilibrium. From rigorous study of that equilibrated structure, further admissible structures follow inevitably in classical and quantum physics, foremost the uniquely admissible dynamical law.

A Linear Metric Contraction Audit

A Mixed-Channel Vector Contractions

Write

$$A_i := h^{jk} C_{ijk}, \qquad B_i := h^{jk} C_{jik}, \qquad D_i := h^{jk} C_{jki}.$$

Proof. For $\mathcal{W} = \Lambda^2 V^* \otimes N$, any $O(V)$-equivariant linear map $\mathcal{W} \to V^*$ is obtained by contracting one copy of C_{ijk} with the metric. Up to relabeling, the only candidates are A_i, B_i, and D_i. Antisymmetry in (i, j) gives $B_i = -A_i$, and the cyclic identity $C_{ijk} + C_{jki} + C_{kij} = 0$ gives $D_i = A_i$. Hence every natural vector contraction is a scalar multiple of A_i, so

$$\dim \mathrm{Hom}_{O(V)}(\mathcal{W}, V^*) = 1.$$

□

B Gauss-Channel Scalar Contractions

$$h^{ik}h^{jl}R_{ijkl} = \mathrm{tr}_h(\mathcal{G}), \qquad h^{ij}h^{kl}R_{ijkl} = 0, \qquad h^{il}h^{jk}R_{ijkl} = -h^{ik}h^{jl}R_{ijkl}.$$

Proof. For $\mathcal{R}$, any $O(V)$-equivariant linear map $\mathcal{R} \to \mathbb{R}$ is a full metric contraction of one algebraic curvature tensor. Up to relabeling, the only possibilities are the three displayed contractions. The middle one vanishes by antisymmetry, and the last equals the first up to sign. Hence every natural scalar contraction is a scalar multiple of scalar-curvature contraction, so

$$\dim \mathrm{Hom}_{O(V)}(\mathcal{R}, \mathbb{R}) = 1.$$

□

C No Mixed-to-Scalar Contraction

$\mathrm{Hom}_{O(V)}(\mathcal{W}, \mathbb{R}) = 0$: $\mathcal{W}$ has odd covariant degree; it cannot be fully contracted with the metric alone.

Proof. Every tensor in $\mathcal{W} = \Lambda^2 V^* \otimes N$ has odd covariant degree. A scalar contraction built only from the metric would have to pair off all covariant indices, which is impossible. Therefore

$$\mathrm{Hom}_{O(V)}(\mathcal{W}, \mathbb{R}) = 0.$$

□

D No Gauss-to-Vector Contraction

$\mathrm{Hom}_{O(V)}(\mathcal{R}, V^*) = 0$: a single copy of R yields either a full scalar or an even-rank tensor; it cannot produce a 1-form.

Proof. A vector output from one algebraic curvature tensor would require leaving exactly one covariant index unpaired after metric contraction. This is impossible: metric contractions remove indices in pairs, so one obtains either a scalar or an even-rank tensor. Therefore

$$\mathrm{Hom}_{O(V)}(\mathcal{R}, V^*) = 0.$$

□

B Mixed-Sector Supplement

A Microlocal Kernel

Definition B.1 (Microlocal mixed kernel subbundle)**.** *Let $S \subset \pi^*(\mathrm{Sym}^2 T^*M \otimes N)$ over S^*M be defined by*

$$S_{(x,\xi)} := \{\xi \odot \xi \otimes \nu : \nu \in N_x\}.$$

Proposition B.2 (Microlocal kernel classification)**.** *For $n \geq 2$, S is a smooth rank-k $O(n) \times O(k)$-equivariant vector subbundle of $\pi^*(\mathrm{Sym}^2 T^*M \otimes N)$, and*

$$S_{(x,\xi)} = \ker \sigma_\xi(L) \subset \mathrm{Sym}^2(T_x^*M) \otimes N_x$$

for every $(x, \xi) \in S^*M$.

Proof. The fiberwise identity is Theorem IV.4. Smoothness and $O(n) \times O(k)$-equivariance follow from the tensorial formula $S_{(x,\xi)} = (\xi \odot \xi) \otimes N_x$. □

Lemma B.2.1 (Gauge-image exclusion for frozen mixed slot)**.** *At a flat compatible background with $K_0 = 0$, the subspace $S_{(x,\xi)}$ from Definition B.1 has trivial intersection with $\operatorname{im} \sigma_\xi(\mathbf{G})$ in the mixed slot, where $\mathbf{G}$ is the infinitesimal internal normal gauge map of Definition III.2.*

Proof. By Lemma IV.3.1, the full mixed-slot gauge linearization vanishes at $K_0 = 0$, so its principal symbol projection to the mixed slot also vanishes. Hence no nonzero vector in $S_{(x,\xi)}$ lies in $\operatorname{im} \sigma_\xi(\mathbf{G})$ in that slot. □

B Canonical Quadratic Invariant

Work locally modulo exact forms:

$$\rho, \rho' \in \Omega^n(M), \qquad \rho \sim \rho' \iff \rho - \rho' = d\beta \text{ for some } \beta \in \Omega^{n-1}(M).$$

Theorem B.3 (Quadratic Minimality)**.** *In form degree 2, with canonical pairing fixed, the unique nontrivial quadratic invariant density is $-\operatorname{tr}(\mathcal{F}^\oplus \wedge *\mathcal{F}^\oplus)$ up to constant.*

Proof. Since $\mathcal{F}^\oplus$ is $\mathfrak{so}(E)$-valued, the linear trace candidate $\operatorname{tr}(\mathcal{F}^\oplus)$ vanishes identically. For a metric connection, invariant polynomials in the curvature are generated by the Pontryagin forms $\operatorname{tr}((\mathcal{F}^\oplus)^{\wedge k})$, so the first nontrivial local invariant order is quadratic. With the canonical pairing fixed, the degree-2 class is therefore unique up to scale and is represented by the canonical contraction

$$-\operatorname{tr}(\mathcal{F}^\oplus \wedge *\mathcal{F}^\oplus),$$

. □

C Quadratic Irreducibility

Definition B.4 (Quadratic decoupling)**.** *A quadratic invariant density in the canonical class is* decoupled *if, after block decomposition*

$$\mathcal{F}^\oplus = \begin{pmatrix} \mathcal{G} & -\mathcal{C}^* \\ \mathcal{C} & \mathcal{S} \end{pmatrix},$$

it is pointwise equal to a density depending only on $(\mathcal{G}, \mathcal{S})$ and not on $\mathcal{C}$.

Theorem B.5 (Quadratic Constraint Obstruction)**.** *Any local quadratic gauge-invariant density built from the fixed canonical pairing is a constant multiple of $-\operatorname{tr}(\mathcal{F}^\oplus \wedge *\mathcal{F}^\oplus)$. In block form, the mixed sector enters with fixed coefficient 2:*

$$\|\mathcal{F}^\oplus\|^2 = \|\mathcal{G}\|^2 + 2\|\mathcal{C}\|^2 + \|\mathcal{S}\|^2.$$

Therefore no nontrivial quadratic invariant suppresses mixed dependence unless $\mathcal{C} \equiv 0$.

Proof. Theorem B.3 gives uniqueness of the quadratic invariant class up to scalar. The displayed norm identity is the block expansion from Proposition II.6. The mixed term is $2\|\mathcal{C}\|^2$ with fixed positive coefficient. Hence suppression of mixed dependence is impossible unless $\mathcal{C} \equiv 0$. □

Data Availability

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

References

[1] I. Kolář, P. W. Michor, and J. Slovák, *Natural Operations in Differential Geometry*, Springer, 1993.

[2] H. Goldschmidt, "Integrability Criteria for Systems of Nonlinear Partial Differential Equations," *J. Differential Geom.* 1 (1967), 269–307. DOI: 10.4310/jdg/1214428094.

[3] S. Kobayashi and K. Nomizu, *Foundations of Differential Geometry, Vol. II*, Wiley-Interscience, 1969.

[4] L. Hörmander, *The Analysis of Linear Partial Differential Operators I*, Springer, 2nd ed., 1990.

[5] M. E. Taylor, *Partial Differential Equations III: Nonlinear Equations*, Springer, 2nd ed., 2011.

[6] M. A. Shubin, *Pseudodifferential Operators and Spectral Theory*, Springer, 2nd ed., 2001.

[7] R. Bartnik and J. Isenberg, "The Constraint Equations," in *The Einstein Equations and the Large Scale Behavior of Gravitational Fields*, Birkhäuser, 2004, pp. 1–38.

[8] Y. Choquet-Bruhat, *General Relativity and the Einstein Equations*, Oxford University Press, 2009.

[9] D. Maxwell, "A class of solutions of the vacuum Einstein constraint equations with freely specified mean curvature," *Math. Res. Lett.* 16 (2009), no. 4, 627–645. arXiv: 0804.0874.

[10] A. Besse, *Einstein Manifolds*, Springer, 1987.